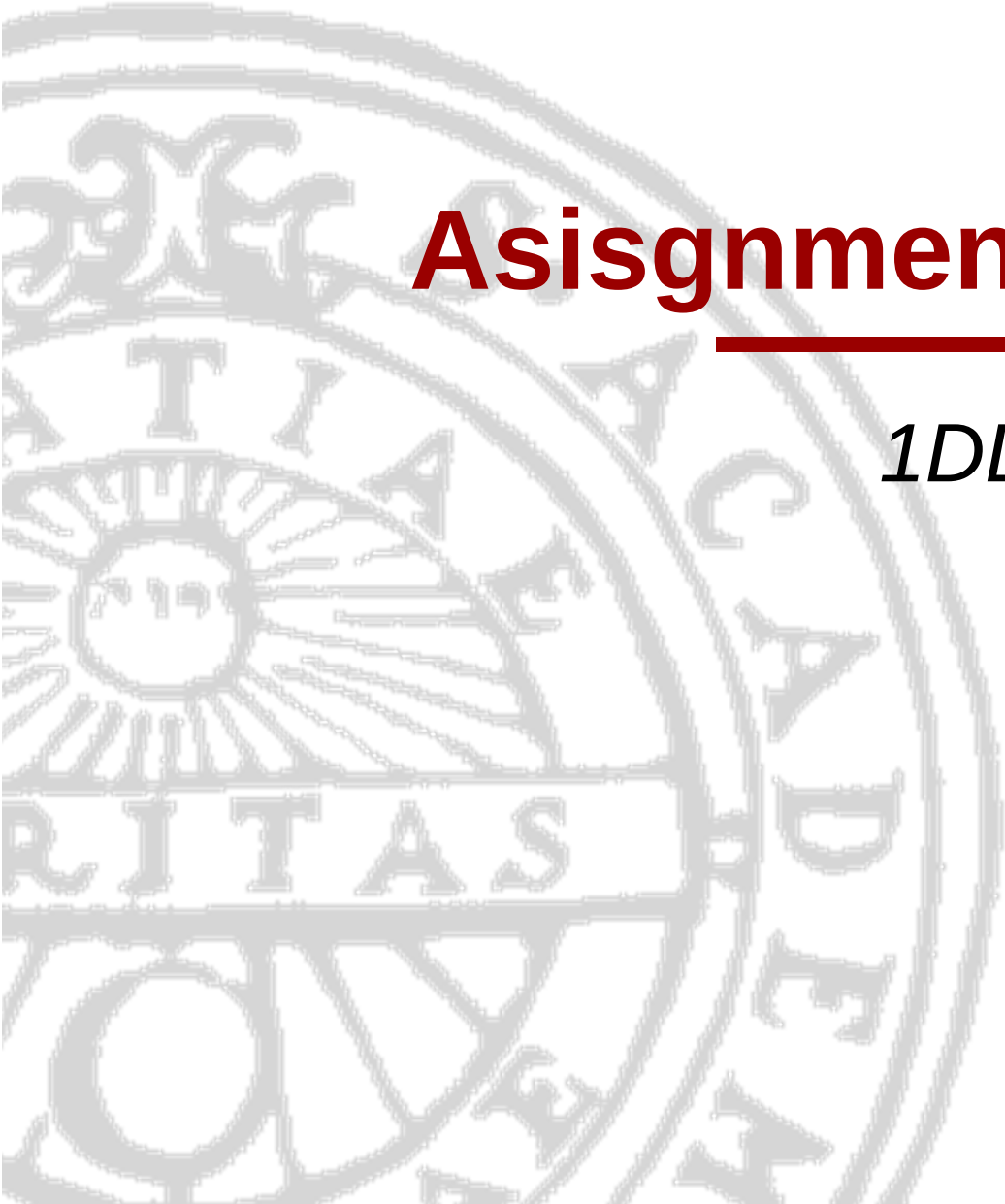




Assignment III - Tutorial

1DL400 – Spring 2012

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UDBL

In assignment 3

- Investigating execution plans of a query
 - ✱ Without index
 - ✱ With index
- Adding KD-tree index implemented in Java to Amos II
 - ✱ Manually adding KD-tree index
 - ✱ Transparently adding KD-tree index using MEXIMA
- Comparing execution plans



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Outline

- Indexing in Amos II
- Extending Amos II with new indexing structures through
 - ✱ Foreign functions in Java
 - ✱ MEXIMA subsystem



Indexing in Amos II

```
■ Database WineSample in which each WineSample has an id, and a feature vector
/*Each wine sample has an identifier*/
create function wsId(WineSample ws)-> Number id as stored;

/*and a feature vector*/
create function features(WineSample ws) -> Vector of Number
featureVector as stored;

/*Define a HASH index on the function feature on second argument
*/
create_index("features", "featureVector", "HASH","multiple");

/*List all indexes on 'features' function.*/
indexes('#'features');
{#[OID 1348 "P_WINESAMPLE.FEATURES->VECTOR-NUMBER"],0,"hash","unique"}
{#[OID 1348 "P_WINESAMPLE.FEATURES->VECTOR-NUMBER"],1,"hash","multiple"}
/*Drop index */
drop_index("features", "featureVector ");
```



Amos II foreign functions in Java

- Foreign function (FF) = User Defined Function (UDF)
- Example

```
/*In AMOSQL to calculate sum of a, and b via a foreign  
function*/
```

```
create function myplus(Number a, Number b) -> Number c as  
foreign "JAVA:FFExample/myplus".
```

```
/*In file FFExample.java*/
```

```
public void myplus(CallContext cxt, Tuple tpl) {  
    ...  
}
```

- Conventions

- ❖ Language : Java
- ❖ File : FFExample.java
- ❖ Function : myplus
- ❖ Path to FFExample.java must be included in CLASSPATH. This is done for you in the setup of assignment 3 (if you prepare the USB stick according to instructions)
- ❖ Implementaion of a FF always has two agruments, namely **cxt**, and **tpl**.
- ❖ Datatype of function is **void**



Amos II foreign functions in Java (cont)

- CallContext - **cxt**

- ❖ A data structure to pass information about the caller.

- Tuple - **tpl**

- ❖ Input and output arguments
- ❖ The order of parameters in **tpl** is the same as in the AmosQL function signature.
- ❖ Example:

`create function myplus(Number a, Number b) -> Number c as
foreign "JAVA:FFExample/myplus".`

a	b	c
Position 0	Position 1	Position 2

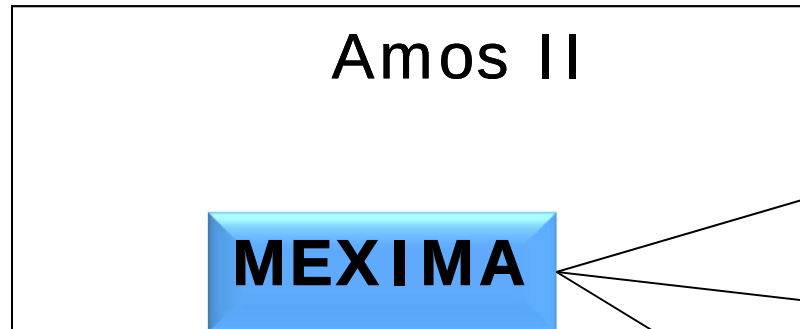
- Some useful functions

- *int Tuple.getIntElem(int pos)*
- *void Tuple.setElem(int pos, int i)*
- *Oid Tuple.getOidElem(int pos)*
- *void Tuple.setElem(int pos, Oid o)*
- *void CallContext.emit(Tuple)* emits a result tuple from a foreign function. Emit is called iteratively when a bag of values is returned.

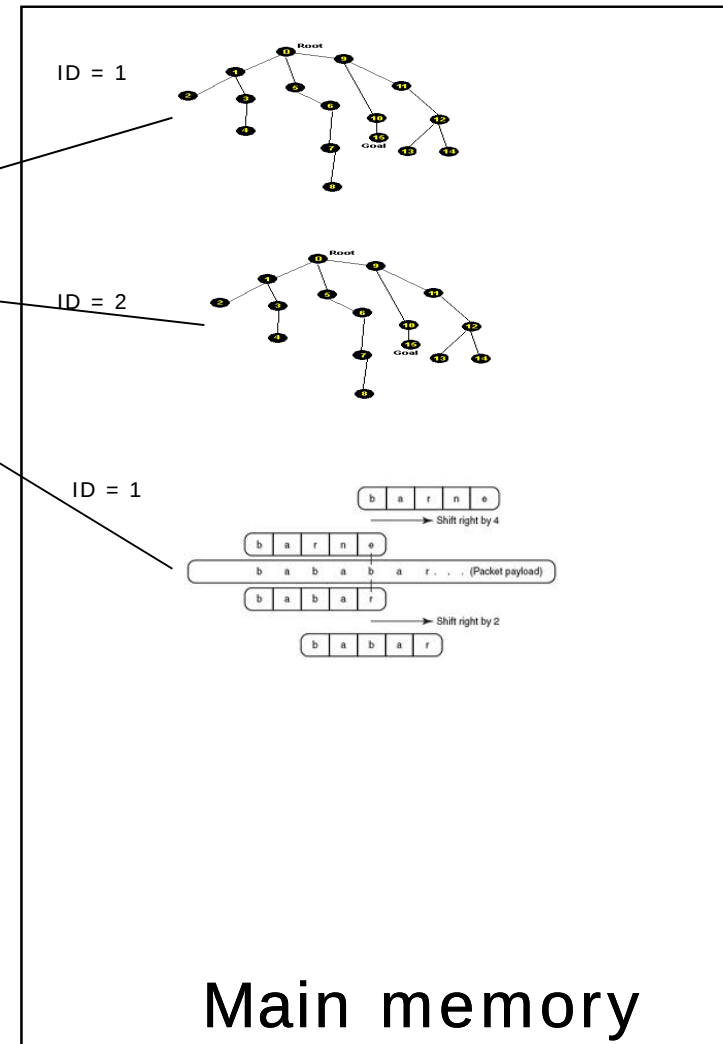


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Integration of new indexing structures

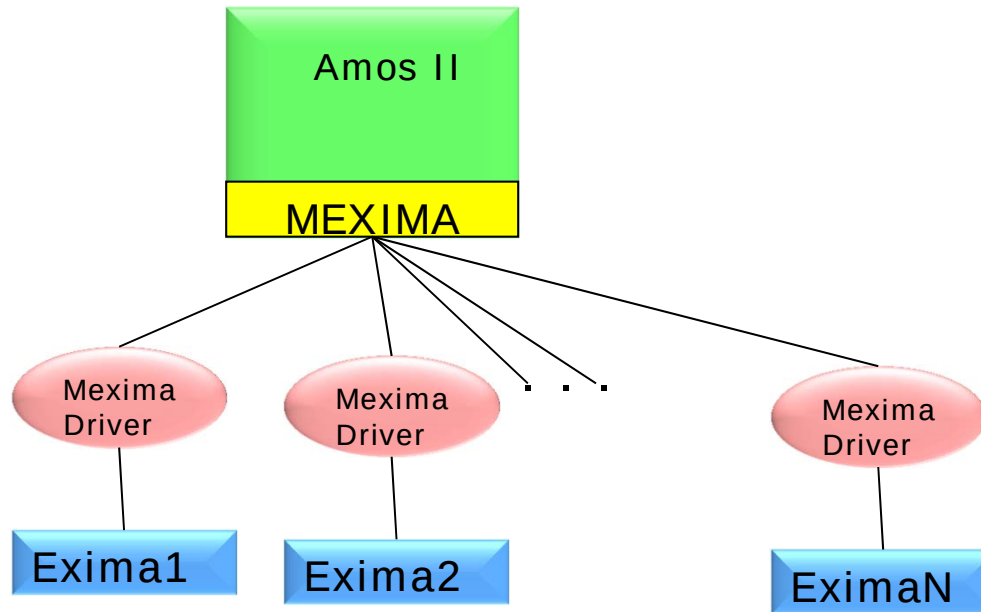


- MEXIMA, a subsystem of Amos, allows including new index structures to the database system.
- Uses foreign functions to interact with external indexing structures.





MEXIMA- Glossary



- + **EXIMA** is an external index manager which are indexing packages outside of Amos.
- + **MEXIMA Driver** implements a set of foreign functions.
- + **MEXIMA** connects Amos II kernel with Exima.

MEXIMA – How to ?

- Implementing a set of foreign functions



OR



- Registering a new external index with Amos
`register_exindextype("indextype");`
- Transparently, using the index type as Amos built-in ones.
 - `create_index, drop_index, . . .`



MEXIMA – Foreign functions

- Operations in which the database system needs to manipulate an index structure are *{ make, get, put, delete, clear, mapper }*.
- We have to implement these operations as foreign function.

- Signatures

```
/*Create a new index of the index type and return its index identifier  
*/
```

```
create function indextype_make() -> Integer as foreign . . .
```

```
/*Store a value val associated with key into an index structure whose  
identifier is id. The value is stored as an Amos object (class Oid)*/
```

```
create function indextype_put(Integer id, Object k, Object val)->Object  
as foreign . . .
```



MEXIMA – Signatures of FF

```
/*Delete a value val associated with key from an index structure whose  
identifier is id.*/
```

```
create function indextype_delete(Integer id, Object f) -> Boolean as  
foreign . . .
```

```
/*Retrieves the value val associated with key from an index structure  
whose identifier is id.*/
```

```
create function indextype_get(Integer id, Object o) -> Bag of Object as  
foreign . . .
```

```
/*Remove the entire index whose identifier is id.*/
```

```
create function indextype_clear(Integer id)->Boolean as foreign. . .
```

```
/*Iterative over the index and emits pairs of keys and values. Note  
that, not all index structures allow iteration over all its data  
items.*/
```

```
create function indextype_mapper(Integer id) -> Bag of Object as foreign  
. . .
```





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Informationsteknologi

DEMO!





Setting up

■ Step 1

- ✱ Download and unzip to **the root directory** of your USB stick from [\[http://udbl2.it.uu.se/Course/DB2_3rdEx.zip\]](http://udbl2.it.uu.se/Course/DB2_3rdEx.zip)

■ Step 2:

- ✱ [Lab] You can run (click) on "assignment3.cmd".
- ✱ [Home]
 - The installed Java version is from or higher than 1.6 ?
 - Edit setup.cmd and change JAVA_HOME to your local Java path.
 - Run (click) on assignment3.cmd.

■ Step 3

- ✱ Compile KDTreeIndex_Stub.java
 - \$ javac KDTreeIndex_Stub.java
- ✱ Run JavaAmos – (*JavaAmos* is a version of the Amos II kernel connected to the Java virtual machine)
 - \$ javaamos



What-to-do ?

- You will modify only
 - ✱ lab3_stub.osql
 - ✱ and KDTreeIndex_Stub.java
- You will have to complete TODO(s) in each file
 - ✱ Modification made on Java file requires
 - Recompiling class file: `$ javac KDTreeIndex_Stub.java`
 - Restarting JavaAmos : `$ javaamos`
 - ✱ **It is good practice to plan/think ahead prior to implementing TODO(s).**



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What-to-hand-in ?

- *Lab3_stub.osql*
- *KDTreeIndex_Stub.java*
- *Group.txt* includes names and emails



Lab resources

■ Amos II

- User manual http://user.it.uu.se/~udbl/amos/doc/amos_users_guide.html
- Tutorial <http://user.it.uu.se/~udbl/amos/doc/tut.pdf>
- Amos II External Interface with Java

■ Complete working example

- Extending Amos II with Hash-Table in Java
- Download from Assignment 3 section – Assignment page